

Inference labs use the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Fit a two-factor ANOVA and distinguish main effects from interactions.
- Visualise an interaction with `emmip()` and read the picture before the table.
- Report a factorial analysis in the correct order: interaction first, then conditional main effects.

Prerequisites

One-way ANOVA.

Background

A factorial design crosses two (or more) categorical factors. The two-way ANOVA decomposes variability into main effects and the interaction between factors. If the interaction is negligible, main effects can be reported as averages across levels of the other factor; if not, the main-effect table is misleading and conditional (simple) effects must be reported instead.

The canonical small dataset is `ToothGrowth`, which crosses dose of vitamin C with delivery method (orange juice vs ascorbic acid). It is a complete factorial and nicely balanced, which means the classification of variance into Type-I, Type-II, and Type-III sums of squares does not matter. With unbalanced designs it does, and Type-III via `car::Anova` with contrasts set to `contr.sum` is the usual safe choice.

Plotting the interaction is almost always the right first step. An interaction plot that shows parallel lines agrees with an insignificant interaction term; crossed lines signal an interaction; lines with different slopes signal a quantitative interaction.

Setup

```
library(tidyverse)
library(broom)
library(car)
library(emmeans)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

Does delivery method alter the effect of dose on tooth length?

Null: no interaction between `supp` and `dose`. Alternative: the dose effect differs by `supp`.

2. Visualise

```
ggplot(tg, aes(dose, len, colour = supp, group = supp)) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  stat_summary(fun = mean, geom = "line") +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.1) +
  labs(x = "Dose (mg/day)", y = "Tooth length", colour = "Supp")
```

3. Assumptions

```
par(mfrow = c(2, 2))
plot(fit)
par(mfrow = c(1, 1))
```

4. Conduct

```
Anova(lm(len ~ supp * dose, data = tg), type = 2)

emm <- emmeans(fit, ~ supp | dose)
pairs(emm)

emmip(fit, supp ~ dose, CIs = TRUE)
```

5. Concluding statement

Tooth length responded to dose (large main effect) and to supplement (smaller main effect), with evidence of an interaction (see ANOVA table). The difference between orange juice and ascorbic acid was largest at the lowest dose and negligible at the highest.

Common pitfalls

- Reporting main effects without first checking the interaction.
- Using Type-III sums of squares with default treatment contrasts in unbalanced designs; the output depends on the contrast coding.
- Ignoring unequal cell sizes when reading an ANOVA table.

Further reading

- Fox J, Weisberg S. *An R Companion to Applied Regression*, ch. 5.
- Cochran WG, Cox GM. *Experimental Designs*.
- Wilkinson L, Rogers CE (1973), *Symbolic description of factorial...*

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)